Mortality Rates among Opioid Addicts in a Longitudinal Study

GEORGE W. JOE, EDD, AND D. DWAYNE SIMPSON, PHD

Abstract: Mortality rates and survival curves were estimated for a sample of opioid addicts who had survived to an initial six-year follow-up interview. A total of 52 of the 555 addicts died between the six-year and 12-year follow-up interviews, which corresponds to an average of 13.8 deaths/1,000 person years. This rate is 6.9 times greater than the general population, adjusting for age. Approximately 29 per cent of the deaths were due to violence and 48 per cent to drugs. (*Am J Public Health* 1987; 77:347–348.)

Introduction

This report represents a 6-year extension of previous research on mortality among opioid addicts who had been treated in community treatment programs.[1,2] It was based on data obtained as part of a 12-year follow-up study of addiction careers.[3,4] Our purpose is to determine the mortality rate in a sample of addicts who had survived until the previous six-year follow-up interview and to determine causes of death. A mortality rate of 13/1,000 person years was reported for opioid addicts in the NIDA-Drug Abuse Reporting Program (DARP),[1] and a six-year follow-up study of opioid addicts after DARP treatment found a rate of 15.2/1,000 person years.[2]

Method

Sample

There were 697 addicts from 18 agencies located across the United States selected for the 12-year DARP follow-up study. Black and White males were randomly selected from methadone maintenance, therapeutic community, outpatient-drug free, outpatient detoxification, and intake-only groups, included in a prior six-year study.[5] Black and White females were selected only from the methadone maintenance group, due to the small number of females in other treatment groups.

The field work for the project was carried out under subcontract by the National Opinion Research Center at the University of Chicago. Of the 697 in the target sample, 555 (80 per cent) were located (70 per cent were interviewed, 2 per cent refused to be interviewed, and 8 per cent were confirmed to be dead). The status of the remaining 142 (20 per cent) could not be determined. Because of this, the mortality rates calculated in the present study are biased. We believe, however, that the bias is small since the nonlocated addicts did not differ substantially from located addicts in terms of demographic measures, pre-DARP behavioral measures, background indicators, and six-year post-DARP outcomes.

Upon admission to the DARP during 1969–72, the average age was 24.9 years; 83 per cent were males and 52 per cent Blacks (and 48 per cent Whites). Besides being addicted to opioids, 55 per cent also used nonopioids daily in the two months before DARP admission, and 83 per cent had been arrested before admission. Between leaving DARP treatment and the six-year follow-up interview, these subjects averaged .83 arrests/year, were employed an average of 51 per cent of the time based on months at risk, and drank an average of 4.5 ounces of 80-proof alcohol/day. Use of drugs during this time averaged less-than-weekly for opioids, monthly for nonopioids, and less-than-weekly for marijuana; the percentages of nonusers for opioids, nonopioids, and marijuana for the six-year follow-up period were 30 per cent, 49 per cent, and 37 per cent, respectively.

Outcomes

The present report examines the mortality rates in relation to age.* Two statistics typically have been used to represent the mortality rate: 1) a crude mortality rate which is computed as the number who die for 1,000 person years of time; and 2) the percentage of the cohort who die per unit of time (usually a year). Both of these statistics are calculated in the present study. The latter statistic is based on the nonparametric estimate (Kaplan-Meier estimator) of the survival curve; specifically, this procedure involves multiplying the natural logarithm of the final Kaplan-Meier estimate by negative one and then multiplying this quantity by 1200/months, where "months" is the number of months corresponding to the Kaplan-Meier estimate used. This nonparametric estimate stems from the following relationship:

$$\int_a^b h(t)\,dt = -\ln S(t)$$

where,

h(t) is the hazard function (i.e., the time-specific mortality rate),

ln S(t) is the natural logarithm of the survival function,

t is the time argument, and

$\int_a^b h(t)\,dt$ is the cumulative hazard function from time a to time b.

By replacing S(t) by the Kaplan-Meier estimator and assuming that the hazard rate is constant over the interval b-a, the nonparametric estimate for the hazard rate can be computed.

The cause of death was classified as violence-related, drug-related, or other.[2] "Violence" includes homicide, gunshot wounds, automobile accidents, hanging, etc. "Drug-related" causes include deaths attributed to overdose or chronic drug abuse, including alcoholism, cirrhosis, hepatic coma, hepatitis, etc. "Other" causes refer to those that could not be classified in the previous two categories (including heart attacks, cancer, stroke, accidents, etc.).

Results and Discussions

Mortality Rates

There were 52 deaths between years 6 and 12 after DARP in the follow-up sample of 555 addicts. The number of deaths per 1,000 person years was 13.8 and the average annual risk

From the Behavioral Research Program, Texas A&M University. Address reprint requests to D. Dwayne Simpson, Director, Behavioral Research Program, and Professor, Department of Psychology, Texas A&M University, College Station, TX 77843. This paper, submitted to the Journal July 2, 1986, was revised and accepted for publication August 5, 1986.

© 1987 American Journal of Public Health 0090-0036/87$1.50

*An analysis of mortality rates in relation to other demographic variables, aspects of deviancy, drug use (including alcohol consumption), employment, and measures that could reflect depression has also been completed, [Joe GW, Simpson DD: Mortality rates and risk factors among opioid addicts in a 12-year follow-up study. College Station, TX: Texas A&M University, Behavioral Research Program, (unpublished) 1986] using Cox's regression analysis.[6]

TABLE 1—Mortality Rates (per 1,000 person years) and Risk of Death per Year

Predictor Variable	N	Mortality Rate	Risk of Death per Year
Total Sample	555	13.8	1.39
Age (years) at DARP Admission			
Under 21	254	10.8	1.15
21–30	128	12.9	1.23
Over 30	171	19.2	2.09

TABLE 2—Kaplan-Meier Estimates of Survival Curves

			Age (years) at DARP					
	Total Sample		Under 21		21 to 30		Over 30	
Month	K-M	S.E.	K-M	S.E.	K-M	S.E.	K-M	S.E.
1	.998	.001	1.000		.992	.008	1.000	
6	.985	.005	1.000		.960	.017	.988	.008
12	.980	.006	.996	.004	.952	.019	.976	.011
18	.972	.007	.992	.006	.944	.020	.965	.014
24	.965	.008	.988	.007	.944	.020	.947	.017
30	.960	.008	.980	.009	.944	.020	.941	.018
36	.954	.009	.972	.010	.937	.021	.941	.018
42	.951	.009	.968	.011	.937	.021	.936	.019
48	.945	.010	.960	.012	.937	.021	.930	.019
54	.942	.010	.956	.013	.937	.021	.924	.020
60	.938	.010	.956	.013	.937	.021	.912	.022
66	.929	.011	.948	.014	.929	.023	.901	.023
72	.922	.011	.936	.015	.929	.023	.895	.023
78	.914	.012	.932	.016	.929	.023	.877	.025
84	.904	.013	.921	.018	.918	.025	.871	.026
N	554		254		128		172	

Note: Persons with missing predictor information were excluded from this table.

of death was 1.39 per cent (95% confidence interval was .84, 1.98 per cent). Mortality rates (per 1,000 person years) and the percentage dying per year for categories of age are summarized in Table 1. Kaplan-Meier estimates of the survival curves for these categories are presented in Table 2.

The rate of 13.8 deaths/1,000 person years is slightly lower than the rate of 15.2 deaths/1,000 person years computed at the six-year follow-up. The overall death rate in the general population adjusted for the DARP age distribution is 2.0/1,000 person years.[7] Thus, the addicts in the present study died at approximately 6.9 times the rate of those in comparable age groups in the general population.

Comparison of the rates by categories of age between the addict sample and the general population indicated where higher risks may exist. When the sample was categorized into age groups 25–27, 27–36, and 37–69, the mortality rates of 10.8, 12.9, and 19.2, respectively, were found to be considerably higher than those of 1.2, 1.3, and 3.1 calculated for comparable age groups in the general population. Nevertheless, the death rate for addicts in the oldest category is smaller than the death rate of 28.0/1,000 person years found in the corresponding oldest category at the six-year interview.

Causes of Death

Of the 52 deaths, 28.8 per cent were classified violence-related, 48.1 per cent as drug-related,and 23.1 per cent as other. Approximately 77 per cent died due to violence or drugs, which compares with the 72 per cent found previously at six years.[2]

The high proportion of deaths that were classified as due to violence or drugs is similar to previous DARP findings, and suggests that the lifestyle and the environment in which these addicts live may be important factors. As cautioned in an earlier study,[2] interpretations of addict mortality rates should also recognize various community and environmental factors, including the social and physical context in which the addicts exist. Generally these addicts still reside in inner-city neighborhoods with high crime rates and in situations of related high mortality risks factors.

ACKNOWLEDGMENTS

This paper reports one of a series of interrelated studies on addict careers and 12-year follow-up outcomes based on the Drug Abuse Reporting Program (DARP). This work was supported by the National Institute on Drug Abuse (NIDA) DA03419 as part of a longitudinal drug abuse research project which began in 1968. The interpretations and conclusions, however, do not necessarily represent the position of NIDA or the US Department of Health and Human Services.

The authors gratefully acknowledge the advice, assistance, and editorial contributions of Drs. W.E.K. Lehman and S.B. Sells.

REFERENCES

1. Watterson O, Simpson DD, Sells SB: Death rates and causes of death among opioid addicts in community drug treatment programs during 1970–1973. Am J Drug Alcohol Abuse 1975; 2:99–111.
2. Joe GW, Lehman W, Simpson DD: Addict death rates during a four-year posttreatment follow-up. Am J Public Health 1982; 72:703–709.
3. Simpson DD, Joe GW, Lehman W, Sells SB: Addiction careers: etiological factors, treatment effects, and 12-year outcomes. J Drug Issues 1986; 16:107–121.
4. Simpson DD, Joe GW, Lehman WEK: Addiction Careers: Summary of studies based on the DARP 12-year follow-up. NIDA Treatment Research Report DHHS No. (ADM 86-1420). Washington, DC: Government Printing Office, 1986.
5. Simpson DD, Sells SB: Effectiveness of treatment for drug abuse: an overview of the DARP research program. Adv Alcohol Substance Abuse 1982; 2:7–29.
6. Cox DR: Regression models and life tables. J R Statist Soc 1972; 34:187–220.
7. National Center for Health Statistics: Advance report of final mortaliy statistics, 1983. *In:* NCHS Monthly Vital Statistics Report, 34(6, Suppl. 2) DHHS Pub. No. (PHS)85-1120. Hyattsville, MD: NCHS, Public Health Service, Sept. 26, 1985.